ML PS3

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1 Bayesian Logistic Regression

1.1 the unormalized posterior of $\beta|Y$

$$p(\beta \mid Y; X, \sigma) \propto \left[\prod_{i=1}^{n} \frac{e^{-(1-y_i)x_i^T\beta}}{1+e^{-x_i^T\beta}}\right] \exp\left(-\frac{\|\beta\|^2}{2\sigma^2}\right))$$

,

1.2 Posterior is proportional to $exp(-U(\beta))$

We have:

$$p(\beta \mid Y; X, \sigma) \propto \left[\prod_{i=1}^{n} \frac{e^{-(1-y_i)x_i^T\beta}}{1+e^{-x_i^T\beta}}\right] \exp\left(-\frac{\|\beta\|^2}{2\sigma^2}\right)$$

and

$$\frac{1}{1+e^{-x_i^T\beta}} = e^{-\log(1+e^{-x_i^T\beta})}$$

Substituting and Combining Exponents:

$$p(\beta \mid Y; X, \sigma) \propto \exp\left(-\sum_{i=1}^{n}\left[(1-y_i)x_i^T\beta + \log(1+e^{-x_i^T\beta})\right] - \frac{\|\beta\|^2}{2\sigma^2}\right)$$

$$\underbrace{\left\{\sum_{i=1}^{n}\left[(1y_i)x_i^T\beta + \log(1+e^{x_i^T\beta})\right] + \frac{\|\eta\|^2}{2\sigma^2}\right\}}_{U(\beta)}$$

So,

$$p(\beta \mid Y; X, \sigma) \propto e^{-U(\beta)}$$

1.3 Classify 0 and 1 (e)

We implemented MAP estimation using gradient descent with 5,000 iterations. The regularization parameter was set to $\sigma = 1.0$, and the learning rate was set to $\epsilon = 10^{-5}$.The model correctly classified all 115 test images, which means the zero-one loss is 0.

1.4 Classify 6 and 8 (f)

The accuracy is 0.9821. The zero-one loss is 0.0179. The regularization parameter was set to $\sigma = 1.2$, and the learning rate was set to $\epsilon = 10^{-4}$.

2 PCA

1. 10 Accuracy=98.26%,Time=0.0270s
2. 20 Accuracy=98.26%, Time=0.0376s
3. 30 Accuracy=100.00%, Time=0.0367s

3 Image Augmentation

Table 1: Classification accuracy with different augmentation factors

Experiment	**Original**	**3×**	**5×**
Run 1	92.86%	80.36%	88.39%
Run 2	92.86%	85.71%	84.82%
Run 3	92.86%	89.29%	93.75%
Average	**92.86%**	**85.12%**	**88.99%**

Across three runs, the original dataset achieved 92.86% accuracy, while the 3× and 5× augmented datasets achieved average accuracies of 85.12% and 88.99%, respectively. Image augmentation should improve generalization performance